WORKED SOLUTION

STEP III 2018, Question 8

Pure • Integrals of Fractional Parts

STEP III 2018, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s3-q8

The key idea

The awkward thing in the integrand is x^{-1}, so substitute $y = 1/x$: this sends $[0,1]$ to $[1,\infty)$, which splits naturally into unit intervals. If f has period 1, each unit interval can be shifted back to $[0,1]$, and the leftover rational factors form a telescoping sum. Part (ii) is then a matter of spotting that $\{x\}$ and $\{2x\}$ both have period 1.

Solution

Part (i)

Put $x = 1/y$, so $\mathrm{d}x = -\frac{1}{y^2}\,\mathrm{d}y$; as x runs from 0 to 1, y runs from ∞ down to 1. Also $1 + x = \frac{y+1}{y}$, so

$$I = \int_{\infty}^{1} \frac{\mathrm{f}(y)}{(y+1)/y}\left(-\frac{1}{y^2}\right)\mathrm{d}y = \int_{1}^{\infty} \frac{\mathrm{f}(y)}{y(1+y)}\,\mathrm{d}y.$$

Splitting $[1,\infty)$ into the intervals $[n, n+1]$ for $n = 1, 2, 3, \ldots$ gives

$$I = \sum_{n=1}^{\infty} \int_{n}^{n+1} \frac{\mathrm{f}(y)}{y(1+y)}\,\mathrm{d}y,$$

as required.

Now suppose $\mathrm{f}(x) = \mathrm{f}(x+1)$ for all x, so that $\mathrm{f}(x+n) = \mathrm{f}(x)$ for every integer n. In the nth integral put $y = x + n$, so that x runs over $[0,1]$:

$$\int_{n}^{n+1} \frac{\mathrm{f}(y)}{y(1+y)}\,\mathrm{d}y = \int_{0}^{1} \frac{\mathrm{f}(x)}{(x+n)(x+n+1)}\,\mathrm{d}x.$$

? **Ask yourself** How do we add up infinitely many integrals like this?

Every integral has the same $f(x)$ and range, so (ignoring convergence, as we are told to) we can swap the sum and the integral and add up the rational factors first. By partial fractions,

$$\frac{1}{(x+n)(x+n+1)} = \frac{1}{x+n} - \frac{1}{x+n+1},$$

so the sum telescopes: for fixed $x \in [0,1]$,

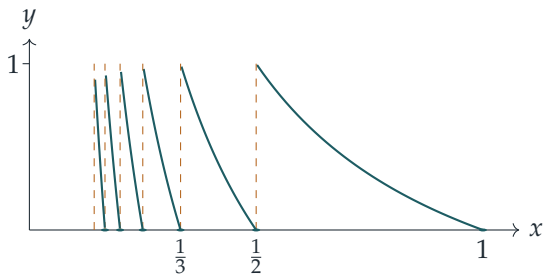
$$\sum_{n=1}^{N}\left(\frac{1}{x+n} - \frac{1}{x+n+1}\right) = \frac{1}{x+1} - \frac{1}{x+N+1} \longrightarrow \frac{1}{1+x} \quad \text{as } N \to \infty.$$

Hence

$$I = \int_0^1 f(x) \sum_{n=1}^{\infty} \frac{1}{(x+n)(x+n+1)}\,dx = \int_0^1 \frac{f(x)}{1+x}\,dx,$$

which is the result we wanted.

Part (ii)



The graph of $\{x^{-1}\}$ above shows why a direct attack is hopeless: on each interval $(\frac{1}{n+1}, \frac{1}{n}]$ it equals $\frac{1}{x} - n$, and there are infinitely many such pieces crowding towards 0. Part (i) avoids all of this.

The first integral. Take $f(x) = \{x\}$. Adding 1 to x adds 1 to $\lfloor x \rfloor$, so $\{x+1\} = \{x\}$ and f has period 1. On $[0,1)$ we have $\{x\} = x$ (the value at the single point $x = 1$ doesn't affect the integral). So by part (i),

$$\int_0^1 \frac{\{x^{-1}\}}{1+x}\,dx = \int_0^1 \frac{x}{1+x}\,dx = \int_0^1 \left(1 - \frac{1}{1+x}\right)dx = 1 - \ln 2.$$

The second integral. Take $f(x) = \{2x\}$, so that $f(x^{-1}) = \{2x^{-1}\}$ as needed.

Ask yourself The function $\{2x\}$ has period $\frac{1}{2}$. Can we still use part (i)?

Yes: anything with period $\frac{1}{2}$ also has period 1, since $\{2(x+1)\} = \{2x+2\} = \{2x\}$. So part (i) applies. On $[0, \frac{1}{2})$, $2x \in [0,1)$ and $\{2x\} = 2x$; on $[\frac{1}{2}, 1)$, $2x \in [1,2)$ and $\{2x\} = 2x - 1$. Therefore

$$\int_0^1 \frac{\{2x^{-1}\}}{1+x}\,dx = \int_0^{1/2} \frac{2x}{1+x}\,dx + \int_{1/2}^1 \frac{2x-1}{1+x}\,dx.$$

Writing $\frac{2x}{1+x} = 2 - \frac{2}{1+x}$ and $\frac{2x-1}{1+x} = 2 - \frac{3}{1+x}$,

$$\int_0^{1/2} \frac{2x}{1+x}\,\mathrm{d}x = 1 - 2\ln\tfrac{3}{2},$$
$$\int_{1/2}^{1} \frac{2x-1}{1+x}\,\mathrm{d}x = 1 - 3\ln\frac{2}{3/2} = 1 - 3\ln\tfrac{4}{3}.$$

Adding, and collecting logarithms,

$$\begin{aligned}\int_0^1 \frac{\{2x^{-1}\}}{1+x}\,\mathrm{d}x &= 2 - 2\ln 3 + 2\ln 2 - 6\ln 2 + 3\ln 3 \\ &= 2 + \ln 3 - 4\ln 2 = 2 + \ln\tfrac{3}{16}.\end{aligned}$$

Checks

- Both answers should lie between 0 and $\int_0^1 \frac{\mathrm{d}x}{1+x} = \ln 2 \approx 0.693$, since $0 \le \{\,\cdot\,\} < 1$. We get $1 - \ln 2 \approx 0.3069$ and $2 + \ln\frac{3}{16} \approx 0.3260$, which are sensible: the fractional part averages about $\frac{1}{2}$, so the answers should be roughly half of 0.693.
- Summing the integrals over the pieces $\left(\frac{1}{n+1}, \frac{1}{n}\right]$ (and halves of them for the second integral) for many thousands of pieces, by computer, gives 0.306853 and 0.326024, matching both answers.

Common slips

- Getting the limits the wrong way round after $y = 1/x$ and losing the minus sign; the minus from $\mathrm{d}x$ and the reversed limits cancel.
- Assuming $\{2x\} = 2x$ on all of $[0,1]$: it drops by 1 at $x = \frac{1}{2}$.
- Thinking part (i) needs period exactly 1; any function with period $\frac{1}{k}$ (for a positive integer k) also has period 1.

Beyond STEP

Swapping an infinite sum with an integral, which we did here without comment, is exactly the kind of step that needs justification at university; theorems such as monotone and dominated convergence say when it is safe (here every term is non-negative when f ≥ 0, which is the easy case). The same substitution idea gives famous integrals of the fractional part, such as $\int_0^1 \{1/x\}\,\mathrm{d}x = 1 - \gamma$, where γ is Euler's constant. For the careful treatment of limits, series and integrals, see Abbott, *Understanding Analysis*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.